Tests & Diagnostics — AGEL0206 σ_e ApJL paper

Last updated: 2026-05-27 (M11 rigorous re-derivation of carried systematics)

Headline (NEW `_mtwdo_` + bad-pixel mask + Balmer-unmasked + He I 3819 mask + M10 full sky-line audit + M11 cube-matched systematic re-derivation + M12 R_e-source fold-in, wide arc-masked window): σ_e(<R_e) = 270 ± 13 km/s = 269.62 ± 13.27 km/s (stat ±5 ⊕ I-shape ±2.27 ⊕ mask ±6.65 ⊕ frame ±5 ⊕ centering ±4 ⊕ window ±3.82 ⊕ reduction ±3.45 ⊕ R_e-source ±6.13) Asymmetric form: 269.62 −13.45 / +13.10 km/s. All headline numbers are emitted deterministically by `scripts/paper_values.py` → `results/PAPER_VALUES.json` (single source of truth; do not hand-edit). Source: `scripts/run_wide_sigma_e.py --cube new_clean_hei`. M11 sweep caches: `results/{ishape_sweep,maskweight_sweep}_wR3800_5400_arcmask_M10/` + `results/nb09a_wavelength_sweep_M10/`. The carried ±15 fit-window component dropped to ±3.82 on the cleaned NEW cube — the dominant source of the ±6 km/s total-budget tightening.

Second-reduction cross-check (OLD cube + bad-pixel mask + Balmer-unmasked + He I + M10 sky masks): σ_e(<R_e) = 262.72 ± 17.99 km/s (asymmetric −18.10 / +17.88). The 6.90 km/s Δ between the two CLEANED + He-I + M10-sky-masked reductions sets the refined ±3.45 km/s reduction-pass systematic (was ±3.86 from clean+He-I-only cubes; M10 sky masks tighten the inter-reduction gap further).

Narrow-window cross-check (Ca H+K + G, OLD cube): σ_e(<R_e) = 268 ± 30 km/s = 267.95 ± 30.10 km/s (stat ±24 ⊕ I-shape ±3.7 ⊕ mask ±7.5 ⊕ frame ±5 ⊕ centering ±4 ⊕ window ±15)

Three-way consistency: the headline (271.87, new clean + He I) and narrow-window cross-check (267.95, old narrow) are within 4 km/s — both inside each other's 1σ stat error. The pre-He-I new-cube wide-window value (268.98) was within 1 km/s of the narrow window; adding the He I mask moves the wide headline ~3 km/s redder of the narrow value, but still well within consistency.

Method (wide, headline): Single ppxf fit on the I-weighted aperture spectrum at R<R_e=2.305" over rest 3800–5400 Å with explicit z=1.302 source-emission masks (MgII λ2800, [OII] λ3727, He I 3819 (M8), [NeIII]/Mg b cluster) + bad-pixel mask + Balmer kept in fit, F200LP spatial arc mask, frame-aware (per-SPS vac/air), SPS-pooled (FSPS+EMILES+XSL) wild bootstrap × polynomial-degree posterior.

Source-emission verification figure (appendix candidate): integrated arc spectrum at `AGEL_0206_ApJL_Figures/AGEL0206_arc_source_spectrum.pdf` — direct extraction from the 69 arc-flagged spaxels (deflector continuum subtracted) showing [OII] 3727 + H9 + [NeIII] + He I 3819 in source-rest frame and the same lines mapped to def-rest (the headline fit window).

Method (narrow, cross-check): Same machinery but on obs-frame 6500–7500 Å (rest 3879–4476 Å, anchored on Ca H+K and G-band only) — the §6cum cumulative I-weighted ppxf path retained from nb07c.

This document is the canonical index of every test, sweep, audit, and sensitivity check run for the σ_e measurement. Each row points to the script/notebook that runs it and the result file or note it produced.

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0. Quick-reference test matrix

#	Test	Where	Result	Status
A. Foundational				

#	Test	Where	Result	Status
A1	KCWI cube provenance — Keck observing logs cross-checked	`NOTES_methodology_2026-04-27.md`, `reference_kcwi_data_properties.md` (updated 2026-05-26)	Verified contributing nights: K409 UT 2025 Aug 30 (12 RED + 4 BLUE DESJ0206 frames) + U002 PI Jones UT 2025 Nov 17 (20 RED + 5 BLUE DESJ0206 frames). Dec 29 2024 RED frames flagged pending FITS-header verification. Sept 29 K409 has zero DESJ0206 entries — earlier note was wrong.	✓
A2	Systemic redshift via line fitting	`notebooks/04_redshift_verification.ipynb`, `scripts/redshift_verify.py`	z = 0.67564 (vs DR2 0.67511)	✓
A3	R_e from masked curve-of-growth (F140W+F200LP)	`scripts/measure_Re.py`, `scripts/final_sigma_e.py:curve_of_growth`	R_e = 2.305" = 16.23 kpc	✓
A4	HST-mean center via centroid_2dg	`scripts/final_sigma_e.py:find_center`	F140W & F200LP agree to 0.36"	✓
A5	Bagpipes SED fitting (M★) — aperture vs Sersic-total	`notebooks/02_streamlined_Bagpipes_SED.ipynb`, `08_sersic_total_photometry.ipynb`	log M★ = 11.33 +0.07/−0.09 (old expert-aperture); 11.40 +0.11/−0.15 (Sersic-total). Superseded by N-series principled masking → headline 11.16 +0.31(sys)	✓
B. Pipeline correctness audits				
B1	Instrumental LSF audit (DISPSCAL=0.294)	`scripts/ifu_spectral_resolution.py`	FWHM=0.692 Å; σ_v_inst≈12-14 km/s	✓
B2	σ_inst sensitivity (×0.5 to ×2.0 LSF)	`scripts/sigma_inst_sensitivity.py`, audit 3	max |Δσ| = 0.83 km/s	✓
B3	SPS template wavelength frames	`scripts/audit_ppxf_methodology.py`, NOTES Test 2	FSPS=vacuum, EMILES=air, XSL=air (Ca K minimum + V_sys)	✓
B4	V_sys air vs vacuum × 5 polynomial degrees	`scripts/audit_ppxf_methodology.py` audit 1	ΔV consistent within ±2 km/s across degrees	✓
B5	z × air-vac differential (obs vs rest application)	`scripts/audit_ppxf_methodology.py` audit 2	1.83 km/s differential — sub-budget	✓
B6	fwhm_gal_dict frame consistency check	`scripts/audit_ppxf_methodology.py` audit 4	dict in REST frame, matches Cappellari pattern	✓
C. SPS templates & frame-fix				
C1	Frame-aware ppxf: per-SPS native frame	`scripts/bootstrap_ppxf.py:SPS_NATIVE_FRAME` + `frame_galaxy='auto'`	V_sys split collapsed from ~110 to ~15 km/s	✓
C2	End-to-end V_sys closure (frame swap)	NOTES Test 3, addendum 2026-04-29	Frame fix matches diagnosis	✓
C3	σ shift from frame fix	NOTES addendum, `error_budget()`	≤5 km/s across SPS → carried as ±5 km/s budget	✓
C4	3-SPS pooled posterior	`scripts/final_sigma_e.py:pool_sps`	Per-SPS V_sys subtracted before pooling	✓
D. Aperture & I(r)				
D1	§6cum cumulative I-weighted ppxf	`scripts/final_sigma_e.py:run_aperture_sps`	σ_e(<R_e) = 267.82 km/s headline	✓

#	Test	Where	Result	Status
D2	§6cum vs §7 (annular) cross-check	`notebooks/07c`, `~/.claude/.../reference_cumulative_vs_annular_sigma_e.md`	§7=255, §7b=271 — all <1σ from §6cum	✓
D3	I(r) shape sweep (10 shapes, fixed mask)	`scripts/run_isource_shape_sweep.py`, `scripts/analyze_isource_shape_sweep.py`	Range = 12.9 km/s (excluding F200LP_Sersic2D outlier) → ±13 budget	✓
D4	I(r) shape sweep refresh post-frame-fix at N=500	NOTES addendum (d), 2026-04-29	Frame fix has <0.5 km/s impact; ±13 budget validated	✓
D5	Aperture choice: R<R_e/8, R<R_e/2, R<R_e	`scripts/final_sigma_e.py:APERTURE_FRACS`	R<R_e/8 dropped (inside seeing FWHM/2 = 0.64”)	✓
D6	Equal-N vs equal-width annular binning	`notebooks/07b` (5-bin), `notebooks/07c` (equal-N)	Equal-N inside R_safe=3R_e/4 chosen	✓
D7	R_e source systematic (mean vs F140W vs F200LP vs CaHK+G)	`scripts/final_sigma_e.py` §7, `notebooks/09` §7b (narrow); `scripts/run_sigma_e_Re_systematic_wide.py`, nb15 (wide)	NARROW: spread = 16.9 km/s (mean=268, F140W=265, F200LP=276, CaHK=282). WIDE (2026-06-08, N=500): spread = 12.25 → ±6.13 km/s (mean 2.305“=269.62, F140W 2.168”=267.44, F200LP 2.441“=272.44, CaHK+G 2.902”=279.69). FOLDED into the M12 budget (was sub-budget only). Light-R_e-family-only spread = ±2.50 (cross-check).	✓ folded (M12)
E. Mask treatment				
E1	F200LP arc mask reprojected to IFU grid	`scripts/final_sigma_e.py:load_setup`	0.7% of all spaxels flagged (~38 inside R<R_e)	✓
E2	Hard mask (w=0.0) headline	`scripts/final_sigma_e.py`, track A	σ_e = 267.82 ± 24 km/s	✓
E3	No-mask sensitivity (w=1.0)	track B	σ_e = 252.82, Δ = −15.0 km/s	✓
E4	Soft-mask single-point (w=0.5)	`scripts/soft_mask_track.py`, track C, NOTES addendum (b)	σ_e = 258.54, Δ = −9.29; super-linear	✓
E5	5-point mask-weight sweep w∈{0,0.25,0.5,0.75,1}	`scripts/soft_mask_track.py --weight ...`, `scripts/analyze_mask_weight_sweep.py`, NOTES addendum (c)	Per-step drops 5.0→4.3→3.5→2.2; quadratic c=+7.3 (concave-up); threshold-dominated	✓
E6	Mask dilation (over-masking diagnostic)	`notebooks/07d_sigma_e_forceful_mask.ipynb`, `~/.claude/.../project_nb07d_overmasking_finding.md`	Dilation inflates σ via noise; do NOT dilate (negative finding)	✓
E7	Arc-spectrum subtraction sibling	`notebooks/07e_sigma_e_arc_subtract.ipynb`	Matches §6cum within 0.1 km/s — residual arc dilution sub-dominant	✓

#	Test	Where	Result	Status
E8	Arc-as-sky mechanism test (no-mask + arc-sky template)	`scripts/run_07f_arc_sky.py`, `notebooks/07f_arc_sky_subtract.ipynb`	σ_D=274.6 vs σ_A=267.8 / σ_B=252.8; recovery=145% — dilution explains the full no-mask Δ	✓
F. Centering				
F1	5-center sweep (±0.4" perturbations)	`NOTES_centering_investigation_2026-04-27.md`, `scripts/regen_s6cum_nomask_diagnostic.py`	σ_e spread = 3.7 km/s → ±4 km/s budget	✓
F2	HST F140W vs F200LP centroid offset	`scripts/final_sigma_e.py:find_center`	0.36" — driven by F200LP arc; HST-mean robust	✓
G. Bootstrap & polynomial degree				
G1	Wild bootstrap (Rademacher × local residual)	`scripts/bootstrap_ppxf.py:run_bootstrap_single_degree`	75-pix rolling window for σ_loc	✓
G2	Parallel bootstrap (joblib, BLAS=1 per worker)	`scripts/bootstrap_ppxf_parallel.py`	~6 min for 6 fits at N=500	✓
G3	Polynomial degree sweep (deg 15-29)	`notebooks/03c`, `scripts/build_nb09.py:§6.5`	σ flat within bootstrap envelope; polynomial saturated	✓
G4	N=50 smoke vs N=500 production agreement	`scripts/final_sigma_e.py --n_bootstrap`	Within 1 km/s	✓
H	Cross-checks (independent estimators)			
H1	§7 discrete Gültekin annular sum (frame-aware, refresh 2026-05-01)	`scripts/refresh_07c_gultekin.py`, `notebooks/07c §7`	256.17 −13.0/+12.7 km/s (post-fix; was 254.99 −24/+28 pre-fix)	✓
H2	§7b flat-σ extrapolation (frame-aware, refresh 2026-05-01)	`scripts/refresh_07c_gultekin.py`, `notebooks/07c §7b`	274.37 −16.2/+17.4 km/s (post-fix; was 271 −33/+35 pre-fix)	✓
H3	F200 mask sensitivity at N=500	`results/annular_bootstrap_07c_nomask/`	Δ_mask = −16.4 km/s → ±16 budget	✓
I. Earlier sigma-discrepancy work (Tests 1-21)				
I1	NB01 vs NB05 σ reproduction	`notebooks/05x` Tests 1-4	Confirmed σ depends on aperture	✓
I2	Source mask + PSF-aware masking	`notebooks/05x` Tests 7-10	F140W mask flags deflector core; F200LP mask is the right arc mask	✓
I3	Threshold sweep + contamination weighting	`notebooks/05x` Tests 11-13	Pre-Gültekin masking strategy work	✓
I4	V_rms profile + bootstrap	`notebooks/05x` Tests 14-16	Pre-Gültekin radial work	✓
I5	Redshift sensitivity (z=0.67511 vs 0.67564)	`notebooks/05x` Test 17	<1 km/s effect on σ	✓
I6	Voronoi binning attempt	`notebooks/05x` Test 18	Failed — abandoned	✓
I7	PowerBin spatial binning	`notebooks/05x` Test 19, `scripts/run_powerbin_test19.py`	Rejected as binning scheme — outer-bin σ>800 km/s	✓
I8	R_e four-way comparison	`notebooks/05x` Test 20, `scripts/measure_Re.py`	F140W+F200LP masked CoG mean = 2.305" headline	✓
I9	Per-spaxel vs PowerBin rotation map	`notebooks/05x` Test 21	No rotation on Re-scale; σ-dominated	✓

#	Test	Where	Result	Status
J. Wide arc-masked window (2026-05-08 → 2026-05-13)				
J1	Wavelength-window sweep (15 windows, narrow → wide)	`notebooks/09a_wavelength_window_sweep.ipynb`, `scripts/run_window_sweep.py`	wR3800_5400 wins on stat precision; XSL template floor at def-rest 3675 Å sets blue edge	✓
J2	Discovery of z=1.302 source-emission contamination	`notebooks/09a §3a`, `notebooks/09b §8`	Bimodal ppxf posterior in wide window (σ≈250 vs 100–150) caused by source emission lines mapped to deflector rest frame	✓
J3	Spectral-mask catalog: 3 source-emission + 1 telluric, def-rest	`scripts/run_window_sweep.py:ARC_MASKS_REST`	Four bands (def-rest 3835-3855 = Mg II z=1.302; 4525-4545 = O_2 A-band telluric edge at obs 7593-7626 Å; 5115-5135 = [O II] z=1.302; 5260-5340 = [Ne III] z=1.302); 212 of 2374 pixels (8.9%) masked. Telluric ID via `NOTES_4534A_spike_investigation_2026-05-18.md`	✓
J4	Arc-mask + ppxf clean=True diagnostic	`notebooks/09b §8`	Bimodality collapses; σ vs polynomial degree FLAT across deg 15-29; clean=True drops 0 pixels	✓
J5	Per-SPS posterior collapse at wide arc-masked	`notebooks/09d §1.3`	SPS spread (FSPS / EMILES / XSL): 26.0 km/s (narrow) → 4.2 km/s (wide arc-masked); SPS systematic essentially disappears	✓
J6	N=500 production at wide arc-masked window	`results/nb09a_wavelength_sweep/wR3800_5400_arcmask_*_N500.npz`	σ_e = 254.9 −7/+5 km/s (stat-only, all-3 SPS pool)	✓
J7	Sersic2D bound-fix (n ∈ [1.0, 6.0])	`scripts/run_isource_shape_sweep.py:181-207`, 2026-05-11	F200LP previously escaped to n=0.30 (unphysical flat-disk), inflating I-shape budget; multi-init grid + tight bounds fix it. I-shape ±5.4 → ±1.5	✓
J8	I-shape sweep at wR3800_5400_arcmask (10 shapes × 3 SPS × N=250)	`results/ishape_sweep_wR3800_5400_arcmask/`	std = 1.5 km/s — 9× tighter than narrow's ±13	✓
J9	Mask-weight sweep at wR3800_5400_arcmask (w ∈ {0, 0.5, 1.0} × 3 SPS × N=250)	`results/maskweight_sweep_wR3800_5400_arcmask/`	peak-to-peak/2 = 3.8 km/s — 4× tighter than narrow's ±16; spectral arc mask absorbs the worst	✓
J10	Three-window cross-check at N=500	nb09 §9: w6500_7500 vs wR3800_5400_arcmask vs wR4000_5400_arcmask	Spread = 15.0 km/s → window systematic ±15 (dominant residual). wR4000_5400_arcmask = orthogonal Hβ + Mg b feature set (no Ca H+K)	✓

#	Test	Where	Result	Status
J11	Both-windows side-by-side budget	`notebooks/09d_final_systematics_both_windows.ipynb`, `results/sigma_e_final_systematics_nb09d.npz`	WIDE 254.85 ± 17.87 vs NARROW 267.95 ± 30.10; consistent at 0.4σ	✓
K. Resolved kinematics (now feasible with wide arc-masked window)				
K1	Per-spaxel ppxf inside R < 1.5 R_e at S/N≥{2,3,5,10}	`notebooks/11_perbin_perspaxel_kinematics_wide.ipynb`	S/N≥5 gives 17 spaxels, $\sigma \in [144, 251]$ km/s with 0 outliers — first clean per-spaxel σ map for this target	✓
K2	PowerBin spatial binning at wide arc-masked window	`notebooks/11 §PowerBin`	7 bins at target S/N=15; 2 outer bins still hit $\sigma > 400$ (irreducible KCWI S/N limit at aperture edge) — improvement vs Test 19 narrow-window failure but not fully cured	⚠ partial
K3	Wide vs narrow window for resolved maps	`notebooks/11`	Wide arc-masked unlocks per-spaxel + PowerBin maps; narrow window was per-spaxel-infeasible	✓
M. Reduction-pass systematic (2026-05-20)				
M1	New `_mtwdo_` red-side reduction PROMOTED to headline 2026-05-26	`notebooks/09e_new_red_reduction.ipynb`, `scripts/run_wide_sigma_e.py --cube new`, `raw_KCWI/New_red/Nov17_2025_DESJ0206_RL_combined_mtwdo_icubes_wcs.fits`	σ_e(<R_e) = 265.76 −18.80 / +18.33 km/s (with new reduction-pass component). Per-SPS spread 2.67 km/s. Δ vs old reduction = +10.92 km/s → carried as ±5.5 km/s reduction-pass systematic (M2). Cube co-aligned to <0.002″ with the old reduction (centering ruled out), shift consistent across all 15 polynomial degrees + all 3 SPS templates (mechanism is distributed across the wide window, not localized to one λ boundary — confirmed by 8000-Å boundary-mask cross-test M3). Caches: `results/run_wide_sigma_e/new/`. Figures: `results/figures/nb09e_reduction_comparison.png`, `nb09e_grating_consistency_check.png`. Audit: `NOTES_nb09e_audit_2026-05-20.md` (Addendum 2026-05-26).	✓ HEADLINE

#	Test	Where	Result	Status
M2	Reduction-pass systematic budget component	M1 vs the old-reduction headline	half-Δ = (265.76 − 254.85) / 2 = ±5.46 km/s, carried as ±5.5 km/s in the systematic budget (matches our peak-to-peak/2 convention from E5 / J10). Flag: only 2 reductions available; revisit / refine if a 3rd lands. Documented in `METHODS_AND_SYSTEMATICS.md` Part I.9 + project memory `project_nb09e_reduction_systematic.md`.	✓
M3	8000-Å boundary-mask cross-test to localize the +10.9 km/s shift	`/tmp/test_8000A_boundary_mask.py`, caches at `results/run_wide_sigma_e/new_8000A_masked/`	σ_e = 268.73 km/s after adding extra mask at def-rest 4715–4834 Å (= obs 7900–8100 Å) → hypothesis REJECTED (shift NOT localized to this one transition). Recovery toward headline = −27%. Confirms the +10.9 km/s shift is distributed across the wide window — likely from the 2-tier per-night flux scaling tilting the entire continuum, not from a single λ boundary.	✓ negative
M4	ppxf clean=True sigma-clip cross-test on new cube (N=100)	`/tmp/test_sigma_clip_new_cube.py`, caches at `results/run_wide_sigma_e/new_sigma_clip_N100/`	σ_e = 265.80 km/s (Δ vs headline = +0.04 km/s). 0 pixels rejected at every (sps, degree) of 45 fits (of ~2161 good pixels each). Reason: ppxf clean=True scales against the noise array; the noise array on this dataset is overestimated, so a 3σ threshold in noise units sits well above the actual outliers (see M5). Test by itself is uninformative; superseded by M5.	✓ negative

#	Test	Where	Result	Status
M5	Local-MAD bad-pixel cleaning (rolling 75-pix window, 3σ threshold) on new cube (N=100)	`/tmp/test_local_mad_clip.py`, caches at `results/run_wide_sigma_e/new_local_mad_clip_N100/`	52 outlier pixels flagged (of 2161 good); biggest is 6-pix cluster at def-rest 5232–5236 Å (obs 8768–8774 Å) at 26σ in local-MAD units — clear unresolved cosmic-ray residual that the noise-scaled clean=True missed. σ_e shifts to 267.18 km/s (Δ vs headline = +1.42 km/s), FSPS-dominated (+3.67). Carried as a stated sensitivity in METHODS Part I.9 ("What is not in the budget"); not folded into formal budget because the shift is well below stat 1σ (±6.1).	✓ sensitivity
M6	OLD-cube replication of M5 clip (same 52 def-rest λ)	`/tmp/test_local_mad_clip_OLD_cube.py`, caches at `results/run_wide_sigma_e/old_local_mad_clip_N100/`	σ_e shifts 254.85 → 256.01 km/s (Δ = +1.16 km/s on OLD cube). Both cubes shift by ~+1.3 km/s → 52 bad pixels are intrinsic to the data, not reduction-specific. Reduction-pass gap preserved (10.91 un-clipped → 11.17 clipped).	✓ replicates
M7	Noise-scaling + ppxf clean=True scan on new cube	`/tmp/test_noise_scaled_clean.py`, caches at `results/run_wide_sigma_e/new_noise_scaled_clean_N100/`	Scanned noise×{1.0, 0.5, 0.3, 0.2, 0.1, 0.05} at FSPS deg=22 → ppxf clips {0, 1, 18, 95, 591, 1247} pixels respectively. Full pipeline at noise×0.3 (19 pix/fit, closest to M5's 52) gives σ_e = 271.04 km/s (Δ = +5.28 km/s) — larger shift than local-MAD because ppxf clean=True is iterative and identifies a different (smaller) set of pixels. Sensitivity envelope: cleaning shifts range +1 to +5 km/s depending on method.	✓ sensitivity

#	Test	Where	Result	Status
M8	He I 3819 source-emission mask added to ARC_MASKS_REST (5237–5253 Å) (2026-05-27)	`scripts/run_wide_sigma_e.py --cube {new,headline}_clean_hei`, caches `results/run_wide_sigma_e/{new,headline}_clean_hei/`	Identified 3-pixel +ve residual cluster at def-rest 5244–5248 Å (= source-rest 3818–3820 Å, matches He I 3819.60 emission from z=1.302 source). Consistent across NEW + OLD reductions → astrophysical, not CR. Adding mask shifts NEW: 268.98 → 271.87 (+2.89), OLD: 260.44 → 264.16 (+3.72). Reduction-pass gap shrinks 8.54 → 7.71 → reduction-pass systematic refined ±4.27 → ±3.86. New TOTAL sys = ±17.25; new paper headline σ_e(<R_e) = 271.87 ± 17.86 km/s (asym −17.99/+17.74).	✓ HEADLINE update
M9	def-rest 5193–5210 Å bump investigation — does NOT mask (it's the Mg b LOSVD wing)	`/tmp/smoke_extra_mask_5190_5210.py` (N=100 smoke); cube-arc spectrum extraction in `AGEL0206_arc_source_spectrum.pdf`; sky-line noise diagnostic	Visible +5–7% bump in deflector data above ppxf fit at def-rest 5193–5204 Å. NOT source emission (arc-spaxel integrated spectrum in source-rest 3782–3788 Å is flat, no emission line — see fig). NOT sky residual (cube `noise_sky` shows 0.6–0.9× median std in this range — well below the strong OH airglow forest at obs 8760–8790 Å that's already covered by BAD_PIXELS + the He I mask). Diagnosis: Mg b LOSVD wing — the bump is mirrored on the blue side of Mg b (def-rest 5160–5170 also has +ve residuals). Mg b at 5183 Å convolved with $\sigma \approx 270$ km/s LOSVD broadens 4–5 Å → wings extend out to ~5190–5210 Å. Smoke test: masking 5190–5210 drops σ_e to 264.83 km/s (−7.04 km/s) — the LARGEST single mask effect seen. Verdict: DO NOT MASK — this region is the *signal* ppxf uses to constrain σ_e, not contamination. Documenting so we don't re-investigate.	✓ DO NOT MASK

#	Test	Where	Result	Status
M10	Full sky-line audit across fit window 3800–5400 Å (2026-05-27)	`BAD_PIXELS_REST` updated to 35 entries (26 original + 9 added). Caches `results/run_wide_sigma_e/{new,headline}_clean_hei/`; per-band rationale inline in `scripts/run_window_sweep.py`. Identification via cube `noise_sky` thresholded at >2.5× median; each added band cross-checked against arc spectrum (`AGEL0206_arc_source_spectrum.pdf`) to confirm no source-emission counterpart.	Found 9 unmasked OH airglow / sky-residual bands in the fit window after user-flagged def-rest ~4600 Å structure: (4380.0-4384.0), (4553.0-4554.6), (4602.0-4610.0) ← the 4600 Å feature, (4687.3-4688.3), (4691.5-4693.6) merged, (4770.8-4771.8), (4951.6-4955.0), (5011.9-5015.3), (5029.8-5033.8). All in def-rest Å, sky_noise 2.5–4× median. NEW cube σ_e: 271.87 → 269.62 (−2.25 km/s). OLD cube σ_e: 264.16 → 262.72 (−1.44 km/s). Reduction-pass gap shrinks 7.71 → 6.90 → refined ±3.86 → ±3.45. New TOTAL sys = ±17.16; new paper headline σ_e(<R_e) = 269.62 ± 17.78 km/s (asym −17.92/+17.65).	✓ HEADLINE update
M11	Rigorous re-derivation of carried I-shape, F200-mask, fit-window systematics on NEW cube + M10 masks (2026-05-27)	Caches `results/ishape_sweep_wR3800_5400_arcmask_M10/` (10 I-shapes × 3 SPS × N=250), `results/maskweight_sweep_wR3800_5400_arcmask_M10/` (3 weights × 3 SPS × N=500), `results/nb09a_wavelength_sweep_M10/` (3 windows × 3 SPS × N=500). Total compute ~1.5 h via `/tmp/run_full_systematics_barrage.sh`.	I-shape: range 266.83–271.37 km/s → peak-to-peak/2 = ±2.27 (was ±1.5 carried). F200 mask: w00=269.69, w50=261.95, w100=256.39 → peak-to-peak/2 = ±6.65 (was ±3.8 carried; bigger because cleaned NEW cube + Balmer-unmasked is more arc-sensitive). Fit-window: wR3800_5400_arcmask=269.66, wR4000_5400_arcmask=268.58, w6500_7500=276.22 → peak-to-peak/2 = ±3.82 (was ±15 carried; cleaned NEW cube + M10 brings wide/narrow into agreement). Net: sys total ±17.16 → ±10.81; TOTAL ±17.78 → ±11.77 (drop of −6.0 km/s). NEW HEADLINE: σ_e(<R_e) = 269.62 ± 11.77 km/s (asym −11.98/+11.57).	✓ HEADLINE update

#	Test	Where	Result	Status
M12	R_e-source fold-in + masking-approach + Hδ cross-checks (2026-06-08, N=500)	`scripts/run_sigma_e_Re_systematic_wide.py` (nb15), `run_sigma_e_mask_systematic.py` (nb13), `run_sigma_e_hdelta_test.py` (nb16); registry `scripts/paper_values.py`→`results/PAPER_VALUES.json`	(1) D7 R_e-source folded in: ±6.13 (wide-window, 4 estimators; user chose full spread 2026-06-08) → sys ±10.81 → ±12.43; HEADLINE σ_e = 269.62 ± 13.27 km/s (asym −13.45/+13.10). (2) Arc-masking-approach systematic = ±5.85 (expert/sersic/perband/global reprojected to IFU grid; spectroscopic analogue of the M★ ±0.16 dex term) — overlaps F200-mask ±6.65, NOT added (larger-of-two). (3) Hδ: KEEP UNMASKED — local-MAD shows Hδ well-fit (max|resid/noise| 0.44 < global 0.81), masking shifts σ_e +6–8 km/s = LOSVD information not contamination (M9 pattern); TODO in `bootstrap_ppxf.py` closed.	✓ HEADLINE update
L. Figure preparation (2026-05-13)				
L1	Figure 2 (narrow) — single posterior inset	`AGEL_0206_ApJL_Figures/figures.ipynb` cell a546db7f, `Mbh_sigma_SED_final.pdf` (was AGEL0206_sigma_e_SED_final.pdf)	Title shows σ_e = 267 ± 24(stat) ± 18(sys) km/s; no-arc-mask overlay removed	✓
L2	Figure 2 (wide, headline) — single posterior inset	figures.ipynb cell `fig2_wide`, `AGEL0206_sigma_e_SED_final_wide.pdf`	σ_e = 255 ± 6(stat) ± 17(sys); residuals panel; 9 absorption features labeled; 4 arc-emission masks shaded	✓
L3	Figure 3 (M_BH-M_star) cleanup	figures.ipynb cell `fig3_clean_no_err`, `Mbh_Mstar_relation_clean.pdf`	No per-object error bars; uniform color for local sample; filtered to Greene+2020 E+S0 list (60 unique, 4 KH13 suffix variants matched; 19 post-KH13 extras dropped)	✓
L4	Figures 2 + 4 updated to principled-mask M★ (2026-05-29)	figures.ipynb cells 19/35/36; `AGEL0206_sigma_e_SED_final_wide.pdf`, `Mbh_Mstar_relation{,_clean}.pdf`	Fig2 inset posterior → perband_raw_10pct (median 11.16) + asymmetric title 11.16 +0.32/−0.08 + shaded fill-in reach to 11.46; Fig4 AGEL point → 10^11.16, asymmetric xerr −0.08/+0.32. Backup `figures.ipynb.bak.Mstar_principled_2026-05-29`	✓

N — Photometry arc-mask verification + principled masking (2026-05-29) | | | | |
N1 | F200LP hand mask reproducible from objective selectors | `scripts/arc_mask_verification.py`,

`notebooks/12` Part I | Color (m_F200LP−m_F140W) + Sersic-residual reproduce the expert F200LP mask; Sersic-residual (k=3) matches expert photometry to 0.016 mag, R_e to 3% | ✓ |
N2 | S/N-regime sweep (over-masking quantified) | `scripts/arc_mask_verification.py`, fig `arcmask_snr_sweep.png` | snr∈{2..20}, k∈{2..8}; F200LP model-flux-fraction stays 0.03–0.10 (clean), Δmag saturates at k≈3 = natural stopping point | ✓ |
N3 | Independent per-band Sersic OVER-MASKS F140W/JWST | `scripts/principled_mask_photometry.py`, figs `principled_*.png` | Single Sersic under-fits bright IR galaxy (median resid +1.0) → over-masks 0.21–0.36 mag; low mdlfrac (0.02–0.03) is a FALSE guard. Use F200LP-reproj instead | ✓ negative |
N4 | Reprojected F200LP footprint = transferable standard | `scripts/mask_method_comparison.py`, figs `maskcompare_*.png` | Reproducing expert JWST mags to 0.01–0.02 mag by reprojecting F200LP (NOT union with F140W — F140W mask covers core, 17px at r<0.4″) | ✓ |
N5 | 2-component (bulge+disk) Sersic model | `scripts/photometry_systematics.py:fit_2comp` | Single Sersic under-fits JWST; 2-comp cuts galaxy-body RMS 3.3–3.5× (F150W2 3.25×, F322W2 3.46×). No PSF (env lacks webbpsf) — flagged | ✓ |
N6 | IR source extension + per-band vs global mask | `scripts/photometry_systematics.py`, figs `photsys_*.png` | Deeper IR extends arc (F150W2 +9119px). raw is mask-size-dependent (up to 0.78 mag per-band↔global); filled is mask-def-independent (≤0.06 mag). Fill-in correction +0.18–0.96 mag | ✓ |
N7 | M★ budget — raw-central, one-sided-up systematic | `results/photometry_systematics_Mstar.npz`, fig `Mstar_budget.png`, `notebooks/12` Part II | log M★ = 11.16 ± 0.08(stat) +0.31(sys) at 10% [11.04 ± 0.14 +0.32 at 20%]; sys = Sersic fill-in (under-arc light), reach 11.46/11.36. per-band vs global negligible for filled. NEW HEADLINE M★ (supersedes 11.33) | ✓ HEADLINE update |
N8 | Explicit masking-approach systematic on M★ | `results/Mstar_masking_systematic.npz`, fig `Mstar_masking_systematic.png` | ±0.16 dex (10%: ±0.160, 20%: ±0.162) = peak-to-peak/2 across {expert-aper, per-band/global × raw/filled} spanning 11.15–11.47. Decomp: under-arc(raw↔filled) ±0.15, mask-def(per-band↔global) ±0.004, mask-extent(expert↔IR-ext) 0.18. Symmetric statement of the headline's one-sided +0.31. TODO: σ_e analogue | ✓ |
N9 | PSF-convolved fill model (firm up the fill leg) | `scripts/psf_fill_model.py`, `results/psf_fill_model.npz` | Gaussian PSF at instrument FWHM (F200LP 0.08″/F140W 0.14″/F150W2 0.06″/F322W2 0.13″; env lacks webbpsf). ΔPSF(filled) ≤ 0.004 mag all bands → ΔM★ ≪0.01 dex, negligible vs stat ±0.08 / masking ±0.16. Fill is PSF-robust (arc outside PSF core) | ✓ |
A3b | R_e pixscale fix in measure_Re.hst_Re (diagnostic) | `scripts/measure_Re.py` (now reads WCS; F200LP literal 0.08→0.05) | F200LP cutout is 0.05″/pix (was hard-coded 0.08) → diagnostic F200LP R_e 3.05→1.91″; F140W unchanged. Headline R_e=2.305″ UNAFFECTED (final_sigma_e.py already WCS-correct: F200LP=2.52″, F140W=2.16″, mean≈2.34″). Flag: two CoG algos differ (measure_Re 1.91 vs final_sigma_e 2.52 for F200LP) — pre-existing | ✓ fix |
A3c | CoG-algorithm reconciliation (measure_Re vs final_sigma_e) | `notebooks/14_CoG_reconciliation.ipynb`, `scripts/measure_Re.py` (`hst_Re` now `r_max_arcsec=6.0` default), `results/Re_cog_reconciliation.npz` | The two CoG algos are the same up to **`r_max`**: measure_Re capped at 4.0″ (`arange(1,80,1)` px), final_sigma_e integrates to 6.0″. Matched r_max → agree <0.04″ (residual = centroid_2dg vs fixed-RA/Dec center). Headline R_e=2.305″ = canonical (final_sigma_e, r_max=6″). New flag: raw CoG R_e is r_max-non-convergent (non-zero sky pedestal); headline sits at the top of a 2.1–2.5″ method family (sky-sub 2.14″, Sérsic 2.18″) → R_e method systematic ±0.08″. Not folded (would shift σ_e aperture + M★; flagged for user). | ✓ reconciled |

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1. Headline numbers

σ_e at three apertures (paper headline = wide arc-masked window)

Aperture	σ_e [km/s] (WIDE, headline)	σ_e [km/s] (NARROW, cross-check)	Notes
R<R_e (=2.305”)	255 ± 18 (254.85 ± 17.87)	268 ± 30 (267.95 ± 30.10)	paper headline — KH13/Greene+20 σ_e at WIDE; NARROW cross-check at 0.4σ
R<R_e/2 (=1.152”)	~225 (nb07c §6cum NARROW)	224 ± 18	gradient diagnostic (still cited from nb07c)
R<R_e/8 (=0.288”)	dropped	dropped	inside seeing FWHM/2 = 0.64”

Error-budget composition (R<R_e) — both windows side by side

Component	WIDE wR3800_5400_arcmask	NARROW w6500_7500	Source
Statistical (bootstrap pooled 1σ)	±6.1	±23.9	wild-bootstrap N=500, FSPS+EMILES+XSL pool
I-shape spread (10 shapes, post-Sersic2D-bound-fix)	±1.5	±3.7	`results/ishape_sweep_wR3800_5400_arcmask/` + `results/annular_bootstrap_07c_ishape/`
F200 mask on/off Δ (peak-to-peak / 2)	±3.8	±7.5	`results/maskweight_sweep_wR3800_5400_arcmask/` + `results/mask_weight_sweep.npz`
Frame fix (max σ shift across SPS, carried)	±5	±5	NOTES addendum + audit 1
Centering (5-center sweep ±0.4”, carried)	±4	±4	`NOTES_centering_investigation_2026-04-27.md`
Fit-window (3-window spread from §9)	±15	±15	nb09 §9: w6500_7500 / wR3800_5400_arcmask / wR4000_5400_arcmask (orthogonal Hβ+Mg b)
Quadrature total	±17.87	±30.10	`results/sigma_e_final_systematics_nb09d.npz`

Where each component changed between windows

- Statistical ±24 → ±6.1 (4× tighter): wide window has ~4× more spectral pixels (2161 vs 555 good pixels). This is the headline reason to elevate the wide window as primary.
- I-shape ±13 → ±1.5 (9× tighter): driven mostly by the Sersic2D bound-fix (J7), which removed the n=0.30 pathology that contributed ±5.4 of the old narrow ±13. The wider window also stabilizes σ across I-shape choices.
- Mask ±16 → ±3.8 (4× tighter): the spectral arc-emission masks (J3) absorb most of what the F200 spatial mask used to absorb; the F200 mask becomes a sub-budget perturbation rather than a primary uncertainty.
- SPS template spread: unstated as a separate line in the budget because it's contained in the statistical pool. But it collapsed 26 → 4 km/s between windows (J5) — was a major budget item at the narrow window, sub-statistical at the wide.
- Frame fix, centering: unchanged (carried from the narrow-window era's careful audits — physically independent of the fit window).
- Fit-window ±15: NEW budget component (didn't exist before nb09d because there was only one window). Spread between three physically meaningful windows: Ca H+K + G (narrow), full canonical range (wide arc-masked), and Hβ + Mg b only (wide minus Ca H+K). This is now the dominant residual systematic.

Sensitivity values (for paper text)

WIDE arc-masked window (headline): - No-spatial-mask (w=1.0) at R<R_e: σ_e = 247.28 km/s — Δ from headline (w=0) = −7.6 km/s - Half-spatial-mask (w=0.5) at R<R_e: 250.15 km/s — Δ = −4.7 km/s - Per-SPS at wR3800_5400_arcmask: FSPS=253.9, EMILES=253.3, XSL=257.5 → spread 4.2 km/s

NARROW Ca H+K + G window (cross-check, retained from nb07c era): - Soft-mask (w=0.5) at R<R_e: 258.5 km/s — Δ from narrow-headline = −9.3 km/s - No-mask (w=1.0) at R<R_e: 252.8 km/s — Δ = −15.0 km/s - σ_e(w) is essentially linear with weak super-linearity (quadratic c=+7.3) - Per-SPS at w6500_7500: FSPS=253.7, EMILES=267.9, XSL=279.6 → spread 26.0 km/s (the SPS systematic that motivated the move to wide)

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2. Detailed test catalog

This expands each row of the matrix above. Subsections reference scripts, notebooks, and result files — open the linked file for the detailed code and the inline plots/output.

A. Foundational data and geometry

A1 KCWI cube provenance — The cube `Nov17_2025_DESJ0206_RL_combined_icubes_wcs.fits` is the final multi-night reduction stacked by K. R. Gupta on UT 2026-01-07. The header's `DATE-OBS = 2025-11-17`, `PROGNAME = U002`, `PROGPI = Jones` describes only the first input frame (`kr251117_00129`), not the full input set.

Verified contributing nights (cross-checked 2026-05-26 against the Keck observing logs):

Night (UT)	Program	PI	DESJ0206 frames	On-source RED + BLUE
2025 Aug 30	K409	TBD	12 RED `kr250830_00090-00101`, 4 BLUE `kb250830_00052-00055`	60 min + 66 min
2025 Nov 17	U002	Jones	20 RED `kr251117_00129-00148`, 5 BLUE `kb251117_00087-00091`	90 min + 110 min
2024 Dec 29	TBD	(Yuguang Chen)	4 RED `kr241229_00092-00095` per local stack list	NOT confirmed as DESJ0206 pointings — only the 44 MB image-only PDF `kcwi-241228-written-log.pdf` (Drive id `16zWO8yaCIvRtoCGenNTndvB7dYdrCkMH`) exists; no machine-readable log. Pending raw-frame header dump.

Confirmed total on-source (Aug 30 + Nov 17): ≈ 170 min RED + 176 min BLUE.

Two earlier-recorded contributors are incorrect: ~~Sept 29 2025 K409~~ has zero DESJ0206 entries in its log; ~~Dec 29 2024~~ remains unverified. Cite K409 + U002 in the paper, flag Dec 29 as pending. See `reference_kcwi_data_properties.md` (memory) for full Drive IDs and log-file pointers.

A2 Systemic redshift — Independent line-by-line Gaussian fitting of Ca H+K, [OII], G-band, etc. gives z_systemic = 0.67564 (NIST air wavelengths). DR2 catalog z=0.67511 (used by older notebooks 03/03b) is offset by ~95 km/s from the line-fit value; nb09 uses 0.67564 throughout. Notebook: `notebooks/04_redshift_verification.ipynb` Module: `scripts/redshift_verify.py` (provides ABSORPTION_LINES / EMISSION_LINES dicts, `gaussian_fit`, `multi_line_fit`, `oii_doublet_fit`)

A3 Effective radius R_e = 2.305" — Mean of F140W and F200LP masked curve-of-growth values (2.168 and 2.441"). The masked CoG zeros mask-flagged HST pixels before the radial integral so the arc/companion contributions don't bias the bright-end half-light point. Computed in `scripts/measure_Re.py` and re-derived in `scripts/final_sigma_e.py:curve_of_growth` for the nb09 pipeline. Cross-check: IFU white-light proper-masked CoG gave 2.61" earlier — the HST-based 2.305" is now the headline.

A4 HST-mean center — `photutils.centroid_2dg` on F140W and F200LP cutouts (with their own `_mask.fits` as the mask) gives sub-pixel centroids in each HST WCS. We average in world coordinates, then propagate to IFU sub-pixel: `scripts/final_sigma_e.py:find_center`. F140W vs F200LP centroid offset = 0.36" — flags the F200LP arc-affected detection but the HST-mean is robust (verified by the 5-center sweep, F1).

A5 Stellar mass — Bagpipes SED fitting on HST + JWST aperture photometry (notebook 02, N_posterior = 500). Bagpipes config: exponential-τ SFH, BPASS+CLOUDY templates, Calzetti dust, free metallicity, redshift fixed at 0.67564.

Per-filter observed flux + best-fit modelled flux:

Filter	λ_pivot [Å]	AB obs	AB model (p50)	F_obs [10^{-18} cgs]	F_model (p50)	residual
F200LP	4 972	22.613	22.563	3.97 ± 0.06	4.16	+4.7%
F140W	13 923	19.134	19.264	12.47 ± 0.05	11.06	−11.3%
F150W2	16 720	18.942	19.033	10.31 ± 0.005	9.49	−8.0%
F322W2	32 470	18.604	18.505	3.73 ± 0.001	4.09	+9.6%

Posteriors:

Parameter	p16	p50	p84
log M★/M⊙	11.24	11.33	11.41
mass-weighted age [Gyr]	3.69	5.10	6.11
exponential SFH age [Gyr]	4.46	5.98	6.97
exponential τ [Gyr]	0.43	0.68	1.21
metallicity Z/Z⊙	0.17	0.89	1.87
A_V [mag]	0.34	0.82	1.55
z_phot	0.674	0.675	0.676

→ M★ = 2.15×10^{11} M⊙; z_phot agrees with the line-fit z = 0.67564 within ±0.001.

The model under-predicts the two NIR bands (F140W, F150W2) by ~8-11% and over-predicts the bluest+reddest by ~5-10%. Classic "redder NIR slope than a single-τ SFH prefers" tension; not catastrophic, but flags that a more flexible SFH (delayed-τ or non-parametric) might tighten the fit if revisited.

Sersic-total cross-check (`notebooks/08_sersic_total_photometry.ipynb`, `scripts/measure_Re.py`-style 2D Sersic fits per filter, extrapolated to total flux):

Filter	AB aperture	AB Sersic-total	Δmag	Sersic n	r_eff ["]
F200LP	22.613	20.672	−1.94	1.42	2.49
F140W	19.134	18.282	−0.85	1.54	1.88
F150W2	18.942	18.154	−0.79	1.40	1.72
F322W2	18.604	17.633	−0.97	1.97	2.03

Re-fed through Bagpipes with the same priors:

Quantity	aperture (paper)	Sersic-total	Δ
log M★/M⊙ p50	11.33 +0.07/−0.09	11.40 +0.11/−0.15	+0.065 dex
M★ [M⊙]	2.15×10^{11}	2.49×10^{11}	+16%

The aperture under-counts outer-profile flux by ~0.8-1.9 mag depending on filter, but at the integrated-mass level the Sersic correction is +0.07 dex — within the aperture-photometry quoted uncertainties. The paper cites the aperture value as the headline; Sersic-total is the sensitivity floor.

Cache: `results/bagpipes_sed_results.npz` (aperture, 500-sample posterior), `results/bagpipes_sersic_refit.npz` (Sersic-total), `results/sersic_total_photometry.npz` (Sersic fit parameters). Figure: `results/AGEL0206_spectra_SED_fit.pdf`.

B. Pipeline correctness audits (`scripts/audit_ppxf_methodology.py`)

Four orthogonal correctness audits run end-to-end on the headline aperture spectrum to verify ppxf usage matches Cappellari (2017, 2023). Saved in `results/ppxf_methodology_audit.npz`. Documented in `NOTES_nb09_frame_fix_and_final_sigma_e_2026-04-28.md` ADDENDUM 2026-04-29.

B1 Instrumental LSF — KCWI header DISPSCAL = 0.294 (instrumental σ in pixels) × CD3_3 = 1.000 Å/pix → FWHM_inst = 2.355 × 0.294 = 0.692 Å (constant in observed Å). At the rest-frame fit window (3879–4476 Å), σ_v_inst ≈ 12–14 km/s. Galaxy σ ≈ 270 km/s is resolved by ~20×. Diagnosed in `scripts/ifu_spectral_resolution.py`.

B2 σ_inst sensitivity — Sweep DISPSCAL × {0.5, 0.75, 1.0, 1.25, 1.5, 2.0}. Max |Δσ| = 0.83 km/s across all SPS. Reason: ppxf's `sps_lib` clips fwhm_diff^2 = (FWHM_gal^2 − FWHM_tem^2).clip(0) (sps_util.py:169). When templates are intrinsically broader (FSPS, EMILES at our band), no convolution is applied — σ becomes insensitive to DISPSCAL. LSF is rock-solid. Script: `scripts/sigma_inst_sensitivity.py`

B3 SPS template frames — Located the Ca K minimum in each shipped template: - FSPS spectra_fsps_9.0.npz: Ca K at 3934.86 Å → +1.20 Å vs air rest = vacuum - EMILES spectra_emiles_9.0.npz: 3933.82 Å → +0.16 Å = air - XSL spectra_xsl_9.0.npz: V_sys closes at +9 km/s with air-conversion → air End-to-end V_sys closure (NOTES Test 3) confirms.

B4 V_sys air vs vacuum × 5 polynomial degrees — Run ppxf at degrees {15, 18, 21, 24, 27} for each SPS in both frames. ΔV(air − vac) ≈ +83 km/s for FSPS, ≈ −82 km/s for EMILES & XSL, consistent within ±2 km/s across degrees. Frame identification is robust.

B5 z × air-vac differential — Apply `vac_to_air` at observed wavelengths (KCWI 6500–7500 Å) → v_offset = −82.7 km/s. Apply at rest wavelengths (3879–4476 Å) → −84.5 km/s. Differential = 1.83 km/s — sub-budget. Either convention is defensible; we follow Cappellari's pattern (apply at obs).

B6 fwhm_gal_dict frame check — Confirmed `_prep_spectrum_for_ppxf` builds the dict as `{"lam": lam_gal_rest, "fwhm": fwhm_gal_rest}` in REST frame, matching `ppxf_example_high_redshift.py:99-101`. FWHM_obs = 0.692 Å → FWHM_rest = 0.413 Å at z=0.67564. `sps_util.py:167` interpolates this onto template `lam_temp` (also rest), then `varsmooth` pre-broadens.

C. SPS templates and frame-fix

C1 Frame-aware ppxf — `scripts/bootstrap_ppxf.py` exposes `frame_galaxy='auto'` which reads `SPS_NATIVE_FRAME = {fsps: vacuum, emiles: air, xsl: air}`. The galaxy spectrum is prepared in the matching frame (`util.vac_to_air()` applied as scalar median when needed, matching `ppxf_example_kinematics_sdss.py` :127-134). Diagnosed Apr 2026 — pre-fix the pipeline always converted to air, producing a −90 km/s FSPS V_sys offset.

C2 V_sys closure — Same galaxy spectrum, swap only the frame. Pre-fix: FSPS −90, EMILES +3, XSL +9. Post-fix (frame-aware): FSPS −7, EMILES +3, XSL +9. Split-track collapsed from ~110 → ~15 km/s. NOTES Test 3.

C3 σ shift from frame fix — Side-effect on σ: ≤5 km/s per SPS, ≤2 km/s on the 3-SPS pool. Carried as ±5 km/s in the error budget (ironically dominated by the per-SPS shifts that pool out, but conservative).

C4 3-SPS pooled posterior — `scripts/final_sigma_e.py:pool_sps` concatenates per-SPS bootstrap σ samples after subtracting per-SPS V_sys medians (so V offsets don't inflate the V posterior; σ posteriors are concatenated raw — no V_sys correction needed for σ). The headline σ_e(<R_e) is the 16/50/84 percentile of this combined-SPS pool.

D. Aperture and I(r) framework

D1 §6cum cumulative I-weighted ppxf — For each spaxel inside R<R_max, build I-weight from the IFU 6500–7500 Å white-light band, drop arc-mask-flagged spaxels, normalize and sum to a single 1-D spectrum, fit ppxf. This is the headline path — matches KH13 / Greene+20 / SAURON / ATLAS3D / MaNGA σ_e definition (Cappellari+2006 eq. 1). Cross-references in `~/.claude/.../reference_cumulative_vs_annular_sigma_e.md`. Code: `scripts/final_sigma_e.py:run_aperture_sps`.

D2 §6cum vs §7 (annular) — §7 discrete Gültekin annular sum at R<R_safe gives 254.99 km/s; §7b flat-σ extrapolation gives 271 km/s; both within 1σ of §6cum's 267.82 km/s. §6cum is the headline because (a) it matches the literature definition, (b) it preserves LOSVD line-shape information that §7's moment-pooling discards, (c) bright-center I-weighting auto-suppresses arc contamination.

D3 I(r) shape sweep (10 shapes) — Hold the spaxel selection fixed (F200-raw mask, R<R_e), vary only the I-weight map. Shapes: 1. IFU_band — 6500-7500 Å (headline) 2. IFU_wl — full IFU white-light 3. F140W_raw — HST F140W reprojected 4. F200LP_raw — HST F200LP reprojected 5. F140W_arcmasked — F140W with arc zeroed 6. F200LP_arcmasked — F200LP with own arc-mask zeroed 7. F140W_1Dcog — annular-mean F140W → interp1d to spaxels 8. F200LP_1Dcog — annular-mean F200LP → interp1d 9. F140W_Sersic2D — full 2D Sersic fit 10. F200LP_Sersic2D — full 2D Sersic fit (poor fit, n=0.30 — outlier)

Range excluding F200LP_Sersic2D outlier = 12.9 km/s → ±13 km/s budget. Script: `scripts/run_isource_shape_sweep.py` Analysis: `scripts/analyze_isource_shape_sweep.py` Caches: `results/annular_bootstrap_07c_ishape/{shape}_{sps}.npz` Figure: `results/figures/nb07c_ishape_sweep.png`

D4 I-shape sweep refresh post-frame-fix — Original sweep at N=250 predated the frame fix. Re-ran at N=500 with `frame_galaxy='auto'` on 2026-04-29. Per-SPS σ shifts <0.5 km/s; spread unchanged at 12.9 km/s; ±13 budget validated. Pre-frame caches preserved at `results/annular_bootstrap_07c_ishape_oldframe/`. NOTES addendum (d).

D5 Aperture choice — Three apertures considered: R<R_e/8 (=0.288"), R<R_e/2 (=1.152"), R<R_e (=2.305"). R<R_e/8 dropped because it sits inside half the seeing FWHM (=0.64") — only 3 spaxels, and the inner σ would underestimate due to seeing-broadened sampling. R<R_e is the paper headline; R<R_e/2 is quoted as a gradient diagnostic.

D6 Equal-N vs equal-width annular binning — For the §7 cross-check only. Equal-N inside R_safe = 3R_e/4 + 1 outer flagged bin (`notebooks/07c`) gives balanced bootstrap variance per bin and isolates arc contamination in the outer bin. Equal-width annuli (`notebooks/07b`, 5 bins) shifted σ_e by ~10 km/s — at the SPS-systematic level (±24). Equal-N is the §7 paper choice.

D7 R_e source systematic (mean vs F140W vs F200LP vs Ca H+K + G) — The headline integrates the I-weighted aperture spectrum inside the mean F140W+F200LP masked CoG R_e (=2.305"). Three alternative R_e definitions test sensitivity to that choice (Track A masked, N=500):

R_e source	R_e ["]	σ_e [km/s]	Δσ_e
mean (paper)	2.305	267.82	—
F140W only	2.168	264.87	−2.96
F200LP only	2.441	275.86	+8.04
Ca H+K + G-band depth	2.866	281.74	+13.92

Spread = 16.9 km/s — at the mask-budget level (±16) but sub-budget vs the total ±32. σ_e rises monotonically with R_e (more outer-bulge spaxels in the I-weighted aperture). The Ca H+K + G-band absorption- depth I-map (rest 3925-3942, 3960-3976, 4297-4313 Å, summed `(continuum − flux)` per spaxel — see `cahk_g_line_depth_map` in `scripts/final_sigma_e.py`) is the most stellar-only of the four; its larger R_e reflects that the deflector light extends further than the F200LP UV-leaning band suggests. Code: `scripts/final_sigma_e.py` §7 + `cahk_g_line_depth_map`. Display: `notebooks/09 §7b` + `results/figures/nb09_re_source_systematic.png`. Caches: `results/final_sigma_e_paper/Re_{F140W,F200LP,CaHK}_{sps}_N500.npz`.

E. Mask treatment

Masking vs no-masking — pro/con summary

Track	Pro	Con
Hard mask (w=0.0, headline)	Removes arc contamination; matches literature convention; F200LP mask is purpose-built for this lens	Drops 38 of 184 spaxels inside R<R_e (~21% by count, ~27% by I-weight) → lower S/N; hard boundary may over-clip a few edge spaxels
No mask (w=1.0, sensitivity)	Maximum S/N; no boundary artifacts	Arc adds low-velocity continuum dilution → σ biased low by 15-17 km/s (Δ_mask = −16 km/s at R<R_e)
Soft mask (w=0.5)	Smooth middle ground	Super-linear in w (quadratic c=+7.3); threshold-dominated bias → first 25% of arc weight already captures 34% of total mask sensitivity. Arbitrary weight choice; not a literature convention
Mask dilation (E6)	—	Inflates σ via noise (dropped spaxels are S/N contributors, not bias contributors). Negative finding — do not dilate

The 16 km/s no-mask Δ is propagated into the error budget as `σ_mask = ±16`, so the choice is bounded, not assumed.

E1 F200LP arc mask reprojection — F200LP `_mask.fits` (HST 0.04“/pix, arc-tuned) is reprojected to the IFU 0.30”/spax grid via `scipy.ndimage.map_coordinates(order=0)` (nearest- neighbor). 69 of 10000 spaxels (0.69%) flagged. Inside R<R_e: 38 of 184 spaxels in the aperture, ~21% — but only ~27% of the I-weight (arc spaxels are mostly off-center).

E2 Hard-mask (w=0.0) headline — track A. E3 No-mask (w=1.0) sensitivity — track B. Δ_mask = −15.0 km/s. E4 Soft-mask single point (w=0.5) — track C. Δ_soft = −9.3 km/s. E5 5-point mask-weight sweep (w∈{0,0.25,0.5,0.75,1}) — Per-step σ drops at R<R_e: 5.0 → 4.3 → 3.5 → 2.2 km/s. Quadratic c = +7.3 (concave-up, super-linear). Threshold-dominated bias: a few heavily-arc-contaminated spaxels carry most of the bias. First 25% of arc weight captures 34% of the total mask sensitivity. Scripts: `scripts/soft_mask_track.py`, `scripts/analyze_mask_weight_sweep.py` Figure: `results/figures/nb09_mask_weight_sweep.png` NOTES addendums (b) and (c).

E6 Mask dilation — Tested whether expanding the F200 mask further reduces σ. Negative finding: dilation inflates σ via noise — the dropped spaxels are S/N contributors, not bias contributors. Memory: `~/.claude/.../project_nb07d_overmasking_finding.md`. Notebook: `nb07d`.

E7 Arc-spectrum subtraction sibling — Build an arc-only spectrum from F200-masked spaxels, fit α × arc + β × deflector model, subtract. Result matches §6cum within 0.1 km/s → *residual* arc dilution (the small fraction that leaks through the F200 mask) is sub-dominant at production statistics. The F200 mask already does the heavy lifting. Note: this test compares “masked + arc-subtract” against “masked alone” — it does NOT directly validate the no-mask vs masked Δ mechanism (that’s E8). Notebook: `nb07e`.

E8 Arc-as-sky mechanism test (rigorous test of the no-mask Δ) — Run §6cum at the *no-mask* R<R_e aperture with the bright outer-arc spectrum passed to ppxf as a `sky` template (free-amplitude additive component, NOT convolved with the deflector LOSVD — physically appropriate since the arc is at z=1.302, decoupled from the deflector kinematics). At N=500, FSPS+EMILES+XSL pooled:

Track	σ_e [km/s]	Δ from σ_A
A: masked headline	267.82	—
B: no-mask	252.82	−15.00
D: no-mask + arc-sky (E8)	274.64	+6.82

Recovery fraction $(\sigma\_D - \sigma\_B)/(\sigma\_A - \sigma\_B) = 145\%$ — i.e. arc-sky overshoots the masked headline by ~7 km/s. Interpretation:

- Continuum dilution explains the FULL no-mask Δ. There is no detectable kinematic-blend or template-mix mechanism contributing in the *opposite* direction.
- The 7 km/s overshoot is consistent with the known caveat that the outer-arc-mask spaxels (R > 3R_e/4) still receive seeing-blurred deflector light at the few-percent level, so subtracting α × arc oversubtracts a small amount of *real* deflector flux too. This inflates σ slightly.
- The masked headline (Track A, paper number) remains the cleaner measurement because it doesn't have this oversubtraction artefact.
- α_arc ≈ 0.31 across all three SPS — internally consistent.

Code: `scripts/run_07f_arc_sky.py` (uses `setup_ppxf_inputs_from_spectrum` + ppxf `sky=` argument). Display: `notebooks/07f_arc_sky_subtract.ipynb`. Caches: `results/arc_sky_07f/{sps}_N500.npz` + `results/arc_sky_07f.npz`. Figures: `results/figures/nb07f_{arc_template,recovery,posteriors}.png`.

F. Centering

F1 5-center sweep — Tested HST_mean (paper choice), F140W only, F200LP only, IFU peak, arc-suppressed IFU peak. σ_e spread = 3.7 km/s. Carried as ±4 km/s in the error budget. Notes: `NOTES_centering_investigation_2026-04-27.md`.

F2 F140W vs F200LP centroid — Δ = 0.36" — driven by the F200LP arc dragging that filter's centroid toward the arc. F140W is the cleaner bulge centroid; HST-mean is the robust compromise.

G. Bootstrap and polynomial degree

G1 Wild bootstrap — Hybrid Rademacher sign-flip × local-residual scaling (75-pix rolling window). For each iteration: residual = (galaxy − bestfit) × sign, scaled by local σ. Re-fit, record V/σ. Implemented in `scripts/bootstrap_ppxf.py:run_bootstrap_single_degree`.

G2 Parallel bootstrap — joblib + threadpoolctl pinning BLAS=1 per worker (otherwise BLAS thread oversubscription pathology — saw 4-hour runtimes). 8 workers gives ~6 min for 6 fits at N=500. Script: `scripts/bootstrap_ppxf_parallel.py:run_bootstrap_single_degree_parallel`.

G3 Polynomial degree sweep — Degrees 15-29 (15 values) per fit. Diagnostic: σ vs degree should be flat within bootstrap envelope; monotonic trend indicates the polynomial is absorbing LOSVD signal. Plotted at `nb09 §6.5` and saved to `results/figures/nb09_sigma_vs_degree.png`. Saturation confirmed at this degree range — picked from earlier `notebooks/03c` analysis.

G4 N=50 vs N=500 — N=50 smoke and N=500 production agree on σ_e to within 1 km/s. N=50 is ~6 min, N=500 is ~50 min. Production headline is N=500; the smoke is for quick rebuilds.

H. Cross-checks

H1 §7 discrete Gültekin (refreshed 2026-05-01) — `notebooks/07c §7`, recomputed with frame-aware ppxf via `scripts/refresh_07c_gultekin.py`. Annular sum $\sigma\_e^2(<R) = \Sigma F\_j (V\_j^2 + \sigma\_j^2) / \Sigma F\_j$ with arc-filter on outer annulus.

	Pre-frame-fix (Apr 24)	Post-frame-fix (May 1)	Δ
σ_e (arc-filtered)	254.99 −24.2/+28.4	256.17 −13.0/+12.7	+1.18

Δ < ±5 km/s frame budget. Errors halved because the V_sys split collapsed from ~99 km/s (pre-fix) to ~8 km/s (post-fix per-SPS V_sys: fsps −11.2, emiles −7.4, xsl −3.1), shrinking the V^2 contribution to the Gültekin sum's MC scatter.

H2 §7b flat-σ extrapolation (refreshed 2026-05-01) — `notebooks/07c §7b`, same refresh. Push the §7 sum into the outer flagged bin assuming σ_outer = σ at the last clean annulus.

	Pre-frame-fix	Post-frame-fix	Δ
σ_e	271 −33/+35	274.37 −16.2/+17.4	+3.37

Both H1 and H2 remain consistent with the §6cum headline (267.82) at <1σ. The point of these cross-checks is preserved by the refresh, and the post-fix bands are tighter, strengthening rather than weakening the consistency.

Note: nb07e (E7, arc-spectrum-subtraction sibling) was *not* refreshed because its result is a relative comparison (matches §6cum within 0.1 km/s) — a uniform per-SPS σ shift moves both arms of the comparison together and the relative match is unchanged.

H3 F200 mask sensitivity (independent) — `results/annular_bootstrap_07c_nomask/`. Same §6cum pipeline with mask off. σ_e(<R_e) = 250.96 km/s. Δ = −16.4 km/s. Independently confirms the nb09 mask-weight sweep at w=1.0.

I. Earlier sigma-discrepancy work (**`notebooks/05x`**)

The 21-test diagnostic notebook resolving the σ ≈ 301 km/s (190-spaxel box, NB01) vs σ ≈ 210 km/s (inner circular aperture, NB05) discrepancy. Resolution: σ depends on aperture; the 190-spaxel box is contaminated by the lensed arc. This work motivated the cumulative-I path (Tests 7-13 explored masking variants, Tests 14-16 went radial, Tests 18-19 tried binning approaches). PowerBin (Test 19) was rejected. The headline analysis moved to circular apertures + cumulative I-weighting (nb06 → nb07c → nb09).

J. Wide arc-masked window (current paper headline, May 2026)

This section documents the May 8–13 work that elevated the rest 3800–5400 Å arc-masked window above the narrow Ca H+K + G-band window as the paper headline. The wide window had been off-limits in the nb07c era because of source-emission contamination from the z=1.302 spiral and template issues — both are now resolved.

J1 Wavelength-window sweep (nb09a) — `notebooks/09a_wavelength_window_sweep.ipynb + scripts/run_window_sweep.py`. Bootstrap σ_e at 15 candidate windows from def-rest 3500–5500 down to narrow Ca H+K only. Result: wR3800_5400 wins on statistical precision (~4× more spectral pixels than narrow). Blue edge set by XSL template floor at def-rest 3675 Å with 5% velocity padding (J2 below); red edge set by KCWI red-channel coverage at z=0.67564.

Provenance of the 3800–5400 range: NOT from a specific literature prescription — it's set by the data (XSL template floor + retaining Ca H+K to cross-anchor with the narrow w6500_7500 result). Closest literature precedent for going as blue as 3800 Å is LEGA-C (Bezanson+2018, van der Wel+2021) at z~0.7 using ~3500–5500 rest. SAURON/ATLAS3D's canonical kinematic window is much narrower (4800–5380; Mg b + Fe5270 + Fe5335 only). The Lick-tradition "4000–5400 (no Ca H+K)" range is captured by the `wR4000_5400_arcmask` sensitivity check (orthogonal Hβ + Mg b feature set) — NOT the headline.

J2 Source-emission contamination discovery (nb09a §3a, nb09b §8) — Wide-window ppxf wild-bootstrap posterior was bimodal (σ ≈ 250 alongside σ ≈ 100–150; frac<200 km/s = 60–80%) at deg ≥ 23. Caused by source-frame emission lines from the z=1.302 spiral arc contaminating the deflector spectrum. Empirically identified as >4σ excess above local continuum:

- def-rest 3835–3855 Å = Mg II λλ2796/2803 doublet from z=1.302
- def-rest 4525–4545 Å = O_2 A-band telluric leading-edge residual at obs 7593–7626 Å (NOT source emission — originally flagged as a ~7.9σ wild-bootstrap excess and provisionally tagged "source rest 3300 Å feature"; relabeled 2026-05-18 after the spike was confirmed present at the same wavelength in deflector R<1.5″, the CLAUDE.md sky box, and a 4–8″ off-deflector annulus, with deflector/off-source amplitude ratio 1.11× despite a 17.5× continuum-brightness ratio. See `NOTES_4534A_spike_investigation_2026-05-18.md` and `results/figures/spike_4534A_diagnostic.png`.)

- def-rest 5115–5135 Å = source [O II] λλ3727/3729 — sits on the Mg b absorption; was distorting the FSPS Mg b depth ratio to 1.18
- def-rest 5260–5340 Å = Mg b / [Ne III] λ3869 cluster from source

Total: 212 of 2374 pixels (8.9% of fit window) masked.

J3 Effect of the arc mask (nb09b §8) — With arc-mask + ppxf clean=True: - σ vs polynomial degree FLAT across deg 15–29 (was bimodal at 250↔175 with a break at deg 23) - N=200 bootstrap collapses to a single sharp peak (frac<200 km/s = 0% across all 3 SPS) - clean=True drops 0 pixels at deg=21 — already clean after arc-mask - σ_e (FSPS+EMILES+XSL pool) = 254.9 −7/+5 km/s at wR3800_5400_arcmask

J4 Per-SPS template spread collapse (nb09d §1.3) —

window	FSPS	EMILES	XSL	all-3 pool	SPS spread
NARROW w6500_7500	253.7	267.9	279.6	268.0	26.0 km/s
WIDE wR3800_5400_arcmask	253.9	253.3	257.5	254.8	4.2 km/s

At the narrow window the 3 SPS libraries fundamentally disagreed by 26 km/s with strong ordering (FSPS < EMILES < XSL — see `reference_sps_systematic.md`). At the wide arc-masked window they agree to 4 km/s, well below per-SPS bootstrap stat error. The SPS-template uncertainty essentially disappears once ppxf has access to Mg b + Fe5270 and the broader feature set. Strong paper argument for elevating the wide window. Figure: `results/figures/nb09d_per_sps_both_windows.png`.

J5 Sersic2D bound-fix (**`scripts/run_isource_shape_sweep.py`**, 2026-05-11) — The I-shape sweep was inflated by a non-physical Sersic2D fit. Original script allowed `n ∈ [0.3, 8.0]` and `ellip ∈ [0.0, 0.95]`. The F200LP fit escaped to n=0.30, ellip=0.00 — a degenerate flat-disk minimum (astropy flagged it as unsuccessful). The flat profile under-weighted the center and yielded σ = 271.8 km/s, a +20 km/s outlier inflating the I-shape std to ±5.4. Fix:

- Tighten bounds: `n ∈ [1.0, 6.0]`, `ellip ∈ [0.0, 0.6]`, `r_eff ∈ [r_eff_pix × 0.5, r_eff_pix × 2.0]` (was ×0.3–×3.0)
- Try 3 initial-condition grids: (n_init, ellip_init) ∈ {(1.5, 0.05), (2.0, 0.2), (3.5, 0.2)}; keep lowest-χ^2 fit
- F200LP now converges to a physical n that gives σ = 251.4 km/s
- F140W was already converging at n=2.14, fix did not change it materially

After the fix, I-shape std collapsed: ±5.4 → ±1.5 km/s at wide arc-masked. Universal fix applied to the I-shape sweep code path — all downstream sweeps now use these bounds.

J6 I-shape sweep at wR3800_5400_arcmask (10 shapes × 3 SPS × N=250) — `results/ishape_sweep_wR3800_5400_arcmask/{shape}_{sps}_N250.npz`. Ten I(r) weight maps: IFU_band, IFU_wl, F140W/F200LP × {raw, arcmasked, 1Dcog, Sersic2D}. Std across the 10 shapes, pooled across 3 SPS = 1.5 km/s. N=100 → N=250 changed result by 0.05 km/s — fully converged.

J7 F200LP spatial-mask sensitivity at wR3800_5400_arcmask (3 weights × 3 SPS × N=250) — `results/maskweight_sweep_wR3800_5400_arcmask/{w00,w50,w100}_{sps}_N250.npz`. w=0 (hard mask) = 254.84, w=0.5 = 250.15, w=1.0 (no spatial mask) = 247.28. Peak-to-peak / 2 = 3.8 km/s. 4× tighter than narrow's ±16 km/s budget because the spectral arc-emission masks now absorb the worst of the arc contamination — the spatial mask is no longer the primary defense.

J8 Three-window cross-check at N=500 (nb09 §9) — Final fit-window systematic. Three windows at full N=500:

window	rest pixels	σ_e (all-3 pool)	role
w6500_7500	obs 6500–7500 Å (rest 3879–4476)	269.7 km/s	narrow Ca H+K + G only
wR3800_5400_arcmask	rest 3800–5336	254.8 km/s	HEADLINE
wR4000_5400_arcmask	rest 4000–5400	~265 km/s	orthogonal Hβ + Mg b (no Ca H+K)

Spread = 15 km/s → fit-window systematic = ±15 km/s. The wR4000_5400_arcmask window is the Lick-tradition feature-set cross-check: removes Ca H+K entirely, so any agreement between the 3800-anchored headline and 4000-anchored alternative is a check that Ca H+K is NOT artificially driving the σ_e measurement. The 10 km/s gap between WIDE and wR4000 is the dominant residual systematic.

J9 Both-windows side-by-side budget (`notebooks/09d_final_systematics_both_windows.ipynb`) — Cache-reuse: narrow I-shape at N=500 reused from `annular_bootstrap_07c_ishape/` (only F200LP_Sersic2D and F140W_Sersic2D refit 2026-05-11 with new Sersic bounds); narrow mask-weight at N=500 reused from `mask_weight_sweep.npz`; both N=500 stat pools from `nb09a_wavelength_sweep/`. Total final budget table (also `results/sigma_e_final_systematics_nb09d.npz`):

component	NARROW w6500_7500	WIDE wR3800_5400_arcmask
stat (N=500)	±23.9	±6.1
I-shape (10 shapes, N=250, Sersic2D bound-fix applied)	±3.7	±1.5
F200 mask (peak-to-peak / 2)	±7.5	±3.8
frame (vac/air, carried)	±5.0	±5.0
centering (HST WCS, carried)	±4.0	±4.0
fit-window (3-window from §9)	±15.0	±15.0
TOTAL	±30.10	±17.87

Headline: σ_e(<R_e) = 254.85 ± 17.87 km/s (WIDE) vs 267.95 ± 30.10 km/s (NARROW). Difference 13.1 km/s, well within both budgets at 0.4σ.

K. Resolved kinematics (per-spaxel + PowerBin at wide arc-masked window)

The wide arc-masked window unlocks resolved-kinematics measurements that were previously infeasible at the narrow window. Documented in `notebooks/11_perbin_perspaxel_kinematics_wide.ipynb` and memory `project_nb11_resolved_kinematics.md`.

K1 Per-spaxel ppxf inside R < 1.5 R_e — Per-spaxel ppxf fits at S/N floors of {2, 3, 5, 10}. Results inside R < 1.5 R_e = 3.46” aperture (358 spaxels total):

S/N floor	N spaxels	% σ in [150,350]	median σ	usable?
≥ 10	0	–	–	max per-spaxel S/N is 8.6
≥ 5	17	94%	201 km/s	YES — production map
≥ 3	~76	68%	~240	usable as soft map with outlier flagging
≥ 2	~134	53%	~243	mostly noise-dominated beyond 1 R_e

S/N ≥ 5 is the cleanest result — 17 central spaxels with σ ∈ [144, 251] km/s, 0 outliers, spanning the central ~1 R_e. σ(R) drops with radius — consistent with elliptical bulge morphology. Aperture-edge spatial resolution is fundamentally constrained by KCWI per-spaxel S/N, not ppxf or fit window.

K2 PowerBin spatial binning at wide arc-masked window — PowerBin (Cappellari 2025) at target capacity S/N=15. 7 bins inside R < 1.5 R_e; median σ = 294.5 km/s. 2 of 7 outer bins still hit σ > 400 km/s — irreducible KCWI per-spaxel S/N limit at the aperture edge. Major improvement over Test 19 narrow-window failure (which had σ > 800 outer bins) but not fully cured. Flag outer bins as noise-limited rather than attempting physical interpretation.

K3 Wide vs narrow for resolved maps — At the narrow w6500_7500 window, both per-spaxel and PowerBin attempts failed (Test 19 in nb05x — outer bins σ > 800 km/s, no per-spaxel S/N above floor). At wR3800_5400_arcmask the same machinery works because the per-pixel S/N quadruples (more line-strength signal per spaxel from the broader feature set + arc-emission masking). This is a direct qualitative gain enabled by the methodology change.

Caches & figures (nb11): - `results/nb11_perbin_perspaxel_wide.npz` — per-bin σ/V/χ^2, per-spaxel σ/V/χ^2, bin_map, V_sys - `results/figures/nb11_perspaxel_sn.png` — S/N map + S/N histogram - `results/figures/nb11_powerbin_map.png` — PowerBin bin map + per-bin S/N - `results/figures/nb11_powerbin_kinematics.png` — σ map, V map, σ(R), V(R) - `results/figures/nb11_perbin_perspaxel_sigma_vs_r.png` — combined σ(R)

L. Paper-figure preparation (2026-05-13)

Final figure cleanup for the ApJL submission, in `../AGEL_0206_ApJL_Figures/figures.ipynb`.

L1 Figure 2 narrow (cell **`a546db7f`**) — Updated title to show σ_e = 267 ± 24 (stat) ± 18 (sys) km/s (stat ± sys separately rather than a single combined ±30). Removed the "no arc mask" blue overlay from the inset; left inset now shows a single red posterior. Saves to `AGEL0206_sigma_e_SED_final.pdf`.

L2 Figure 2 wide (cell **`fig2_wide`**, NEW, paper headline) — σ_e = 255 ± 6 (stat) ± 17 (sys) km/s. I-weighted aperture spectrum over rest 3800–5400 Å with 9 absorption features labeled (Ca K, Ca H, Hδ, G-band, Hγ, Hβ, Mg b, Fe5270, Fe5335) and 4 source-emission arc masks hatched. Residual sub-panel underneath. Inset shows single red posterior (no-arc-mask comparison removed). Saves to `AGEL0206_sigma_e_SED_final_wide.pdf`.

L3 Figure 3 M_BH–M_star cleanup (cell **`fig3_clean_no_err`**, NEW) — Cleaned version of the comparison plot. NO per-object error bars, uniform medium-gray color for all local early-types (no E vs S0 / KH13 vs post-KH13 subdivision). Filtered to Greene+2020 E+S0 list (60 unique galaxies): dropped 19 post-KH13 additions that were not in Greene+2020 Fig 10 (mostly Saglia+2016 / Thater+2019 fills), matched 4 KH13 suffix variants (NGC 1316b / 2778c / 3998c / 4486A) to their Greene+2020 names (NGC 1316 / 2778 / 3998 / 4486A). All literature relations (Greene+20, Reines & Volonteri+15, Farrah+25, Sahu+24) preserved. AGEL0206 point at M★=10^11.3, M_BH=6.5e8 preserved. Saves to `Mbh_Mstar_relation_clean.pdf`. The legacy cell 33 with per-object error bars and E vs S0 color subdivision is preserved as `Mbh_Mstar_relation.pdf`.

Sanity / withdrawn: during the wide-window writeup an erroneous "Cappellari 2017 recommends extending blueward" citation was drafted in chat — retracted, no such recommendation exists in the ppxf paper. The wavelength-range choice is data-driven (XSL template floor + retain Ca H+K), with LEGA-C (van der Wel+21) as the closest literature precedent (see J1 above).

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3. File index

Documents (top-level)

File	Purpose
`TESTS_AND_DIAGNOSTICS.md`	This file — master test index
`NOTES_nb09_frame_fix_and_final_sigma_e_2026-04-28.md`	Frame fix derivation + addenda (a)–(d)
`HANDOVER.md`	Apr 27 σ_e analysis snapshot (pre-frame-fix; superseded by nb09)
`HANDOFF_Ir_sweep_2026-04-28.md`	I(r) sweep design doc (consumed; sweep complete)
`NOTES_methodology_2026-04-27.md`	Cube provenance + earlier methodology
`NOTES_centering_investigation_2026-04-27.md`	5-center sweep result + ±4 km/s budget
`CLAUDE.md`	Repo conventions, key results summary
`README.md`	High-level project description

Notebooks (paper-ready / current)

Notebook	Purpose
`notebooks/09_final_sigma_e_paper.ipynb`	Paper-ready σ_e at both windows; §9 three-window cross-check, §10 I-shape N=250, §10.1 mask-weight N=250
`notebooks/09a_wavelength_window_sweep.ipynb`	15-window scan; identifies wR3800_5400 sweet spot

Notebook	Purpose
notebooks/09b_lsf_and_sigma_clip.ipynb	Arc-mask + ppxf clean=True diagnostic with 3×3 zoom plots
notebooks/09d_final_systematics_both_windows.ipynb	Side-by-side WIDE vs NARROW final budget (paper headline writeup)
notebooks/11_perbin_perspaxel_kinematics_wide.ipynb	Per-spaxel + PowerBin σ map at wR3800_5400_arcmask (NEW — previously infeasible)
notebooks/04_redshift_verification.ipynb	$z = 0.67564$ line-fit
notebooks/02_streamlined_Bagpipes_SED.ipynb	$\log M_\star = 11.33$
notebooks/08_sersic_total_photometry.ipynb	Sersic-total photometry cross-check
notebooks/07c_sigma_e_equalN.ipynb	§6cum + §7 equal-N annular cross-check (narrow window, retained for cross-check)
notebooks/07e_sigma_e_arc_subtract.ipynb	Arc-subtraction sibling cross-check
notebooks/05x_sigma_discrepancy_diagnostic.ipynb	Tests 1-21 (foundational diagnostics)

Notebooks (history / reference, not paper-driving)

`01_*`, `02_*`, `03_bootstrap_ppxf_errors`, `03b_*`, `03c_*`, `05_*`, `06_*`, `07_*`, `07a_*`, `07b_*`, `07d_*` — kept for the history of the analysis.

Scripts (production)

Script	Purpose
scripts/final_sigma_e.py	nb09 production driver — runs all 3 mask tracks at N=500
scripts/build_nb09.py	Generates the paper notebook from the saved artifacts
scripts/bootstrap_ppxf.py	Frame-aware ppxf input prep + wild bootstrap
scripts/bootstrap_ppxf_parallel.py	joblib parallel bootstrap (BLAS=1 per worker)
scripts/measure_Re.py	R_e via masked CoG (multi-band, multi-strategy)
scripts/redshift_verify.py	Line-fitting redshift module
scripts/ifu_spectral_resolution.py	LSF audit

Scripts (sweeps and audits)

Script	Purpose
scripts/audit_ppxf_methodology.py	4 orthogonal correctness audits
scripts/sigma_inst_sensitivity.py	DISPSCAL × {0.5..2.0} sweep
scripts/run_isource_shape_sweep.py	10-shape I(r) sweep — Sersic2D bound-fix applied 2026-05-11 ($n \in [1.0, 6.0]$ + multi-init grid)
scripts/analyze_isource_shape_sweep.py	I-shape sweep analysis + figure
scripts/run_window_sweep.py	Wavelength-window driver with arc-mask flag; underpins nb09a + nb09 §9
scripts/soft_mask_track.py	Mask-weight runs (CLI `--weight`)
scripts/analyze_mask_weight_sweep.py	5-point sweep analysis + linearity fit
scripts/regen_s6cum_nomask_diagnostic.py	Centering investigation
scripts/relock_nomask_Re_N500.py	One-off N=500 relock for the no-mask track

Result files (key)

File	Contents
WIDE arc-masked (current headline)	
results/sigma_e_final_systematics_nb09d.npz	Both-windows side-by-side budget (paper headline writeup)
results/nb09a_wavelength_sweep/wR3800_5400_arcmask_*_T*_N500.npz	N=500 stat pool at headline window

File	Contents
results/nb09a_wavelength_sweep/wR4000_5400_arcmask_*_T*_N500.npz	N=500 orthogonal-feature-set cross-check
results/ishape_sweep_wR3800_5400_arcmask/	10 shapes × 3 SPS × N=250 (I-shape ±1.5)
results/maskweight_sweep_wR3800_5400_arcmask/	3 weights × 3 SPS × N=250 (mask ±3.8)
results/sigma_e_window_sweep_09a.npz	15-window posterior summary
results/nb11_perbin_perspaxel_wide.npz	Per-bin + per-spaxel kinematics at wide
NARROW Ca H+K + G (cross-check)	
results/final_sigma_e_paper.npz	nb09 narrow headline + per-SPS + 3-track summaries
results/final_sigma_e_paper/	Per-(aperture, SPS, mask_weight) caches
results/sigma_e_radial_07c.npz	nb07c §6cum + §7 + §7b posteriors
results/annular_bootstrap_07c/	Per-aperture cumulative I-weighted caches
results/annular_bootstrap_07c_ishape/	Per-(shape, SPS) I-shape caches (N=500, F140W/F200LP Sersic2D refit 2026-05-11)
results/annular_bootstrap_07c_nomask/	§6cum nomask caches
results/mask_weight_sweep.npz	5-point mask-weight sweep summary (narrow)
Audits & figures	
results/ppxf_methodology_audit.npz	4-audit results
results/figures/nb09_*.png	nb09 paper figures (incl. fit windows + I-shape comparison)
results/figures/nb09d_per_sps_both_windows.png	Per-SPS posterior collapse (26→4 km/s)
results/figures/nb11_*.png	Resolved-kinematics maps (per-spaxel S/N, σ map, V map, $\sigma(R)$, $V(R)$)
results/figures/nb07c_ishape_sweep.png	I-shape sweep figure
../AGEL_0206_ApJL_Figures/AGEL0206_sigma_e_SED_final_wide.pdf	Paper Figure 2 (wide, headline)
../AGEL_0206_ApJL_Figures/AGEL0206_sigma_e_SED_final.pdf	Paper Figure 2 (narrow, cross-check)
../AGEL_0206_ApJL_Figures/Mbh_Mstar_relation_clean.pdf	Paper Figure 3 (clean, no error bars, Greene+2020-matched)

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4. Reproduction recipes

```
conda activate ISMgas

# === Headline pipeline (~50 min at N=500) ===
python scripts/final_sigma_e.py --n_bootstrap 500
python scripts/build_nb09.py
jupyter nbconvert --to notebook --execute --inplace notebooks/09_final_sigma_e_paper.ipynb

# === Methodology audits (~3 min) ===
python scripts/audit_ppxf_methodology.py
python scripts/sigma_inst_sensitivity.py

# === I-shape sweep (~50 min at N=500) ===
python scripts/run_isource_shape_sweep.py --n-bootstrap 500 --n-jobs 8
python scripts/analyze_isource_shape_sweep.py

# === Mask-weight sweep (~25 min total) ===
# (w=0 and w=1 are produced by final_sigma_e.py; only need 0.25/0.5/0.75)
python scripts/soft_mask_track.py --weight 0.25
python scripts/soft_mask_track.py --weight 0.50
python scripts/soft_mask_track.py --weight 0.75
python scripts/analyze_mask_weight_sweep.py

# === Smoke run (~6 min, useful for rebuilds) ===
```

```
python scripts/final_sigma_e.py --n_bootstrap 50
```

Caches are keyed by N_bootstrap and mask_weight in the filename — re-runs at the same N skip cached fits unless `--force` is passed.

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5. What's intentionally NOT done

Updated 2026-05-13 with status changes after the wide arc-masked window work and the Sersic2D bound-fix.

- Re-run nb09 at higher N (e.g. N=2000): the pooled posterior is already smooth; at the wide arc-masked window the stat error is ±6 km/s while the fit-window systematic dominates at ±15 km/s. More N is wasted compute.
- ~~Re-fit Sersic2D F200LP~~ DONE 2026-05-11 — the n=0.30 fit was a parameter-bound pathology, not a UV-arc artifact. Bound-fix (J7) tightened bounds to n ∈ [1.0, 6.0] with multi-init grid. F200LP_Sersic2D now converges to a physical fit (σ=251.4 km/s) and is kept in the I-shape budget, not flagged as an outlier.
- Voronoi binning — Test 18 / nb05x — abandoned; not pursued at N=500.
- ~~PowerBin spatial binning~~ PARTIAL — Test 19 narrow-window rejected was correct at the time. At the wide arc-masked window (K2 / nb11) PowerBin now works for the inner ~5 of 7 bins; 2 outer bins still hit $\sigma > 400$ (irreducible KCWI per-spaxel S/N at aperture edge). Not used as headline binning scheme but published as supplementary σ map.
- ~~Per-spaxel rotation map at production statistics~~ DONE (K1 / nb11) — at the wide arc-masked window, S/N ≥ 5 gives 17 central spaxels with clean σ values and a visible σ(R) gradient. No coherent rotation detected above the noise (consistent with elliptical-bulge expectation). Published as supplementary kinematic map.
- Full nb07a Sersic-I revisit at the post-frame-fix headline — superseded by the I-shape sweep (D3/D4/J6) which covers the Sersic2D shape with the bound-fix applied.
- Re-quote σ_e on KH13/Bell+2003 stellar masses for the M_BH–M★ plot — Figure 3 keeps the KH13 dual-M/L dynamical bulge masses for the local comparison sample. The Greene+2020 list match (L3) restricts to the same 60 galaxies as Greene+2020 Fig 10 but with our (better) bulge mass values.
- Cross-match nb09a "narrow" window to other AGEL targets — paper is single-target. The arc-emission-mask methodology (J2/J3) is reusable for future AGEL targets where source z is known, but applying it elsewhere is not in scope for the ApJL.

See `~/.claude/projects/.../memory/MEMORY.md` for compressed cross-session references, including: - `reference_cumulative_vs_annular_sigma_e.md` — §6cum vs §7 design choice - `reference_sps_systematic.md` — per-SPS V_sys + frame-fix details - `reference_ppxf_vacair_handling.md` — vac/air conversion methodology - `reference_kcwi_data_properties.md` — KCWI cube + LSF + paper boilerplate - `project_nb09_final_sigma_e.md` — current headline status